Python Programming Foundations

Week 1 - First Programs and Basic Values

Day 1: "What a Python Program Does"

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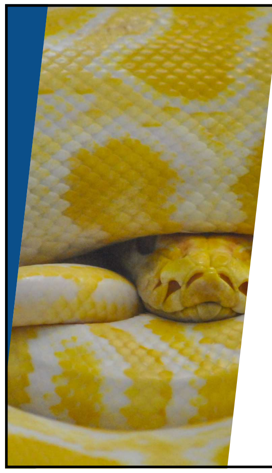

Make Something Happen

Your first goal is to make Python do something visible.

If you can run a tiny program and explain the output, you are doing real programming.

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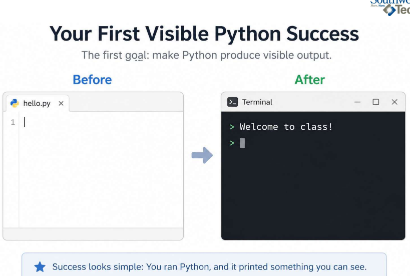

First Program

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Start Small, Build Outward

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You are not expected to know everything today; you are expected to make one

This course starts with small working programs and builds outward.

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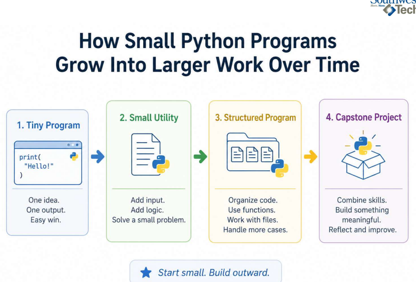

Program Growth

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What Counts as Success Today

- run the program
- change one value
- run it again
- explain what changed

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What We Will Use Today

- `print()`
- strings
- variables
- visible output

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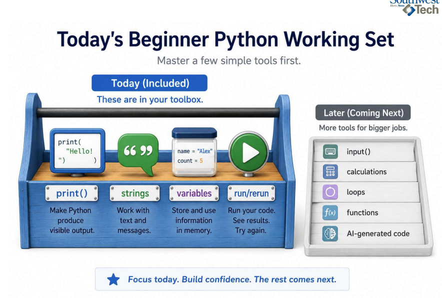

Working Set

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What We Will “Skip” For Now

- `input()`
- Calculations
- Loops
- Functions
- AI-generated code

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A Program Is Instructions

Python follows the instructions in your file.

Most beginner programs run from top to bottom - one line at a time.

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Python Runs Instructions Line by Line

1.
```
course_name = "Python Programming"
```
2.
```
student_name = "Jordan"
```
3.
```
print(student_name)
```

Python runs the next instruction in order.

Line-by-Line Exectuion

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`print()` Makes Output Visible

`print()` makes a value or message appear on the screen.

Visible output is your first evidence that the program ran.

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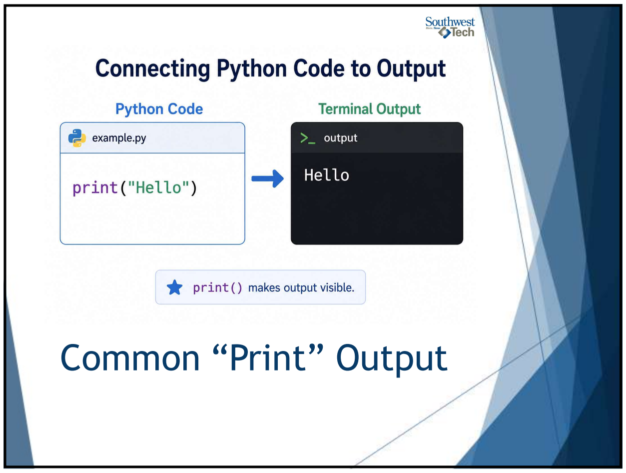

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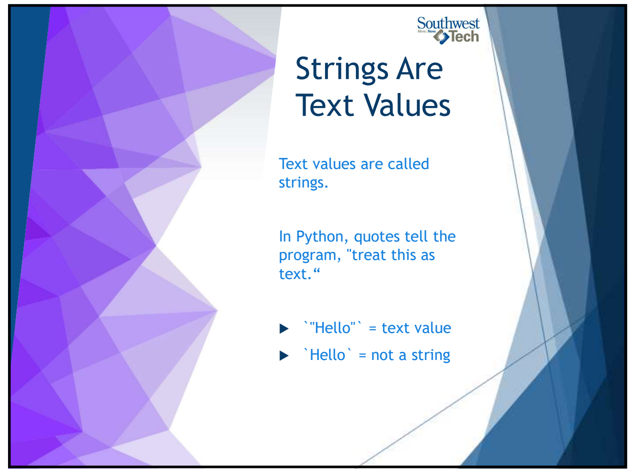

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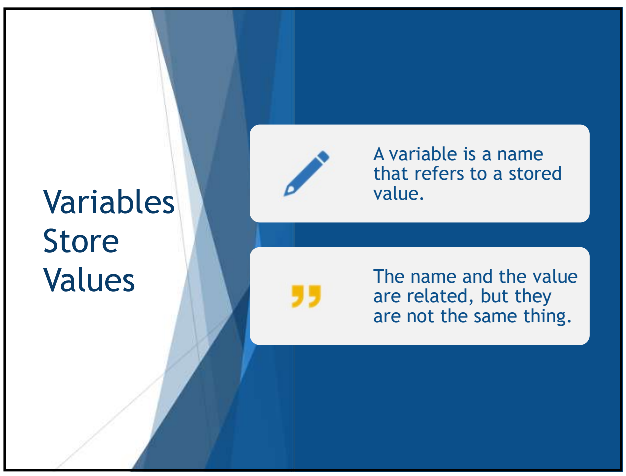

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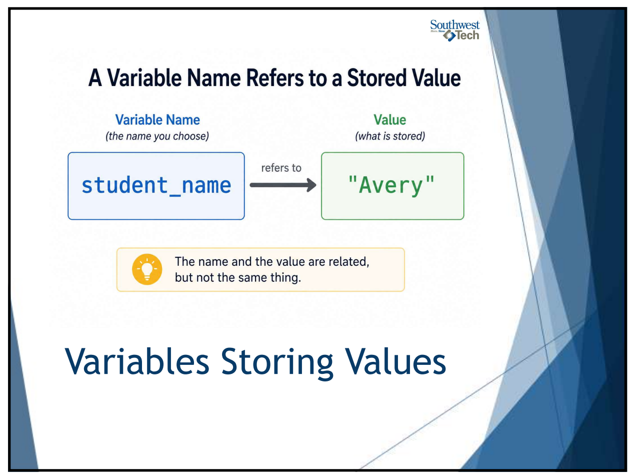

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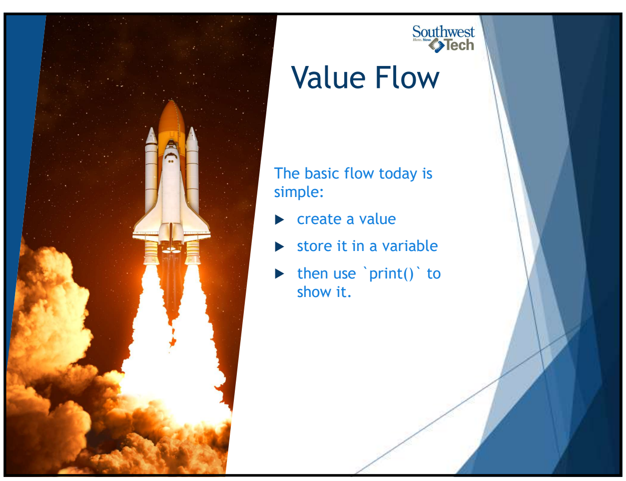

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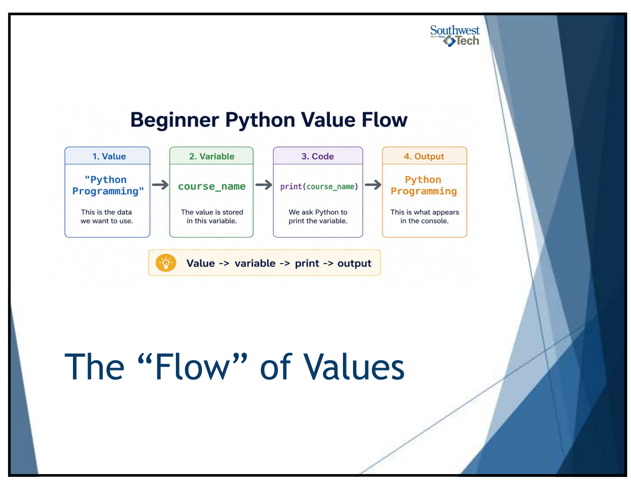

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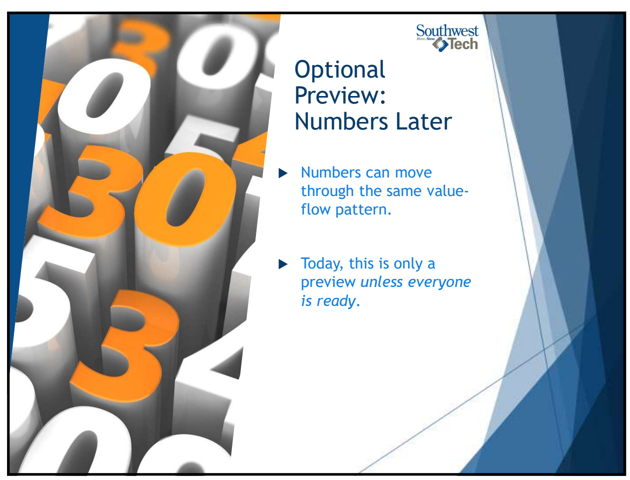

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Common Failure: Missing Quotes

`"Hello"` is a string because it has quotes.

`Hello` without quotes is not automatically text to Python.

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Quotes are *Important*

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Common Failure: Name vs Value

The variable name is how we refer to the value.

Changing the stored value can change what the program prints later.

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From Demo to Lab

In Lab/Assignment 01:

Create one tiny program with a greeting and one variable. Then change the variable value and rerun the program.

1. Create a new `.py` file.
2. Print a greeting.
3. Store one text value in a variable.
4. Print that variable.
5. Change the stored value.
6. Rerun the program.

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Lab Evidence

- Save the `.py` file and make sure it runs.
- Be ready to explain one value you changed and how the output changed.

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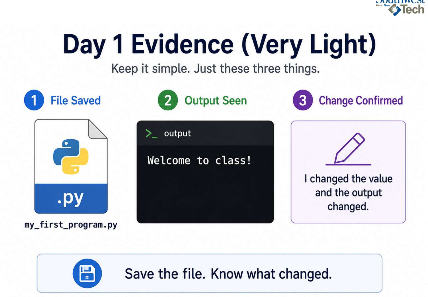

Lab Evidence

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Closing Success Check

- By the end of today, you should be able to run a tiny Python program.
- You should also be able to explain what one variable stores.

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Thank you!

Have Fun!

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